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Data Base Management Systems (25B11CS001)

Capstone Stone Project Documentation



ADITYA UNIVERSITY

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SCHOOL OF COMPUTING

Department of Artificial Intelligence and Machine Learning

CERTIFICATE

This is to certify that the Cornerstone project work entitled "**Recycle Management System**" is submitted by

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in partial fulfillment of the requirements for the award of the B.Tech. degree in Artificial Intelligence and Machine Learning for the academic year 2026-2027.

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DECLARATION

We hereby declare that the Cornerstone project entitled "**Recycle Management System**" is original and is submitted to **ADITYA UNIVERSITY**, Surampalem, in partial fulfillment of the requirements for the award of the **B.Tech. Degree in Artificial Intelligence and Machine Learning**. We further declare that this project work has not been submitted, either in full or in part, for the award of any degree in this or any other institution. We also confirm that this work complies with the Institutional Academic Integrity and AI Usage Policy

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ABSTRACT

The Recycle Management System is a database-driven application designed to centralize and manage recyclable waste records, collection activities, recycling transactions, collection centers, collectors, and reward information. The system addresses the challenge of scattered waste-management records by providing a structured relational database foundation using MySQL.

The database is normalized to reduce redundancy and maintain data integrity through primary keys, foreign keys, unique constraints, NOT NULL conditions, and CHECK constraints. It supports efficient storage and retrieval of waste collection and recycling information and provides a reliable backend for reporting and analysis.

The system enables the recording of waste categories, recyclable materials, collection requests, collection centers, collectors, recycling quantities, processing outcomes, and user rewards. By organizing the data into core entities and associative relationships, the design provides a scalable foundation for monitoring recycling activities.

The project demonstrates the implementation of SQL DDL, DML, and DQL operations, along with joins, aggregation, subqueries, views, updates, deletion, and reporting queries. The system can help organizations track recyclable waste, improve collection coordination, monitor recycling quantities, and support data-driven environmental management.

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ACRONYMS

Acronym	Full Form
DBMS	Database Management System
RMS	Recycling Management System
ER	Entity–Relationship
RDBMS	Relational Database Management System
SQL	Structured Query Language
1:1	One-to-One
1:M	One-to-Many
M:1	Many-to-One
M:N	Many-to-Many
PK	Primary Key
FK	Foreign Key
CRUD	Create, Read, Update, Delete
1NF	First Normal Form
2NF	Second Normal Form
3NF	Third Normal Form
UI	User Interface
DDL	Data Definition Language
DML	Data Manipulation Language

1. INTRODUCTION

1.1 Brief Information about the Project

The Recycle Management System is a DBMS project that establishes a structured database foundation for managing recyclable waste and collection operations. The system centralizes waste items, users, collection requests, collectors, collection centers, recycling transactions, and rewards into a single relational database. MySQL is used as the backend relational database.

The core idea is to replace fragmented registers and disconnected files with a centralized system that supports efficient data storage, retrieval, and analysis. The design connects waste records with collection and recycling activities, enabling administrators to monitor quantities, materials, centers, and recycling outcomes.

1.2 Motivation and Contribution of Project

The project is motivated by several common challenges in waste and recycling management:

- **Scattered Waste Records:** Waste collection details may be maintained in multiple registers or spreadsheets, making tracking difficult.
- **Data Inconsistency**: Repeated manual entries can result in duplicate or inconsistent waste records.
- **Difficult Collection Tracking**: It can be difficult to identify collection requests, assigned collectors, and collection status.
- **Limited Recycling Analysis**: Recycling quantities and material-wise performance are difficult to summarize without a centralized database.
- **Reporting Challenges**: Manual preparation of collection and recycling reports can be time-consuming.

The contribution of this project is a normalized relational database that provides a single source of truth for recyclable waste management and supports efficient reporting and analysis.

1.3 Objectives of the Project

The primary objectives of the project are:

Objective	Description
Store Data Efficiently	Centralized database for recyclable waste and collection data.
Reduce Redundancy	Normalized tables minimize duplicate information.
Improve Data Security	PK, FK, UNIQUE, NOT NULL and CHECK constraints.
Track Collections	Record requests, collectors, centers, quantities and status.
Monitor Recycling	Store collected and processed quantities.
Generate Reports	SQL queries provide useful recycling statistics.

1.4 Scope of the Project

The scope of the project covers the following key areas:

- **User Management:** Store user details who submit recyclable waste for collection.
- **Waste Management:** Record recyclable items, categories, quantities, and conditions.
- **Collection Management:** Create and track waste collection requests.
- **Collector Management:** Maintain collector details and assigned collection activities.
- **Center Management:** Store recycling and collection center information and capacity.
- **Recycling Transactions:** Record collected quantities, processed quantities, and recycling status.
- **Rewards and Reporting:** Maintain reward points and generate useful management reports.

2. SYSTEM ANALYSIS

2.1 Existing System

The existing system for recyclable waste management may suffer from several limitations:

- **Manual Record Maintenance:** Waste and collection information is often maintained in notebooks, Excel sheets, or disconnected files.
- Duplicate **Records:** Repeated entries can lead to inconsistent information about users, waste items, and collections.
- **Time-Consuming Searches:** Finding a particular collection request or recycling transaction can take considerable time.
- **Difficulty in Generating Reports:** Material-wise, center-wise, and collector-wise summaries are difficult to prepare manually.
- **Higher Possibility of Data-Entry Errors:** Manual entry can produce incorrect quantities, dates, or statuses.

2.2 Proposed System

The proposed system uses a centralized relational database to organize recyclable waste information. It employs primary keys, foreign keys, uniqueness rules, NOT NULL conditions, and CHECK constraints to maintain data integrity. The system supports user management, waste categories, waste items, collection requests, collectors, collection centers, recycling transactions, rewards, and reporting.

2.3 Functional Requirements

Module	Functional Requirement
User Management	Add, update, search, and display user details.
Collector Management	Manage collector details.
Collection Request	Record and track requests.
Waste Management	Store waste type and quantity.

Recycling Center	Manage center details and capacity.
Transaction	Record processed quantities.
Search and Reporting	Retrieve records and generate reports.
CRUD Operations	Insert, view, update, and delete records.

2.4 Non-Functional Requirements

- **Performance:** SQL queries should retrieve recycling-related records efficiently.
- **Integrity**: Primary keys and foreign keys should maintain valid relationships between database tables.
- **Security**: Database access privileges should be controlled to prevent unauthorized modifications.
- **Usability**: Database tables and query results should be understandable and easy to use.
- **Reliability:** Database transactions should maintain consistent data during insert, update, and delete operations.
- **Scalability:** The database design should support an increase in users, collectors, collection requests, waste records, and transactions.
- **Maintainability:** The database structure should allow tables and queries to be updated when project requirements change.

2.5 Requirements Specification

Category	Suggested Requirement
Hardware	Intel i3 or above, 4 GB RAM.
Operating System	Windows 10/11
Database	MySQL
Software	MySQL Server or Workbench.

3. DATABASE DESIGN

3.1 Major Entities

The database design includes sixcore entities:

Entity	Description
User	Stores user details, contact information, and address.
Collector	Stores collector details and assigned area.
Collection_Request	Maintains waste collection requests and their status.
Waste	Stores waste type, quantity, and category.
Recycling_Center	Stores recycling center location, capacity, and contact information.
Transaction	Records processed quantities, dates, collectors, and recycling centers.

3.2 ER Model

- **Entity:** A real-world object represented in the database. Example: User, Collector.
- **Attribute:** A property that describes an entity. Example: User_Name, Waste_Type.
- **Primary Key**: Uniquely identifies each record. Example: User_ID.
- **Foreign Key**: Links a record between table. Example: User_ID in Collection_Request.
- **Relationship**: An association between entities. Example: USER → COLLECTION_REQUEST.
- **Cardinality**: Defines the relationship between entities. Example: 1:M, M:1.

3.3 Relationships

The relationships between entities are as follows:

Relationship	Type	Description
User records Waste_Item	1:M	One user can register multiple waste items.
Collection Request contains Waste	1:M	One request, multiple waste records.
Collector handles Collection	1:M	One collector can handle multiple collection requests.
Waste generates Transaction	1:M	One waste record, multiple transactions.
Recycling Center processes Transaction	M:1	One center, multiple transactions.

4. ER DIAGRAM

The ER diagram for the Recycling Management System is shown below. It illustrates the six main entities, their attributes, relationships, and cardinalities used in the database design.

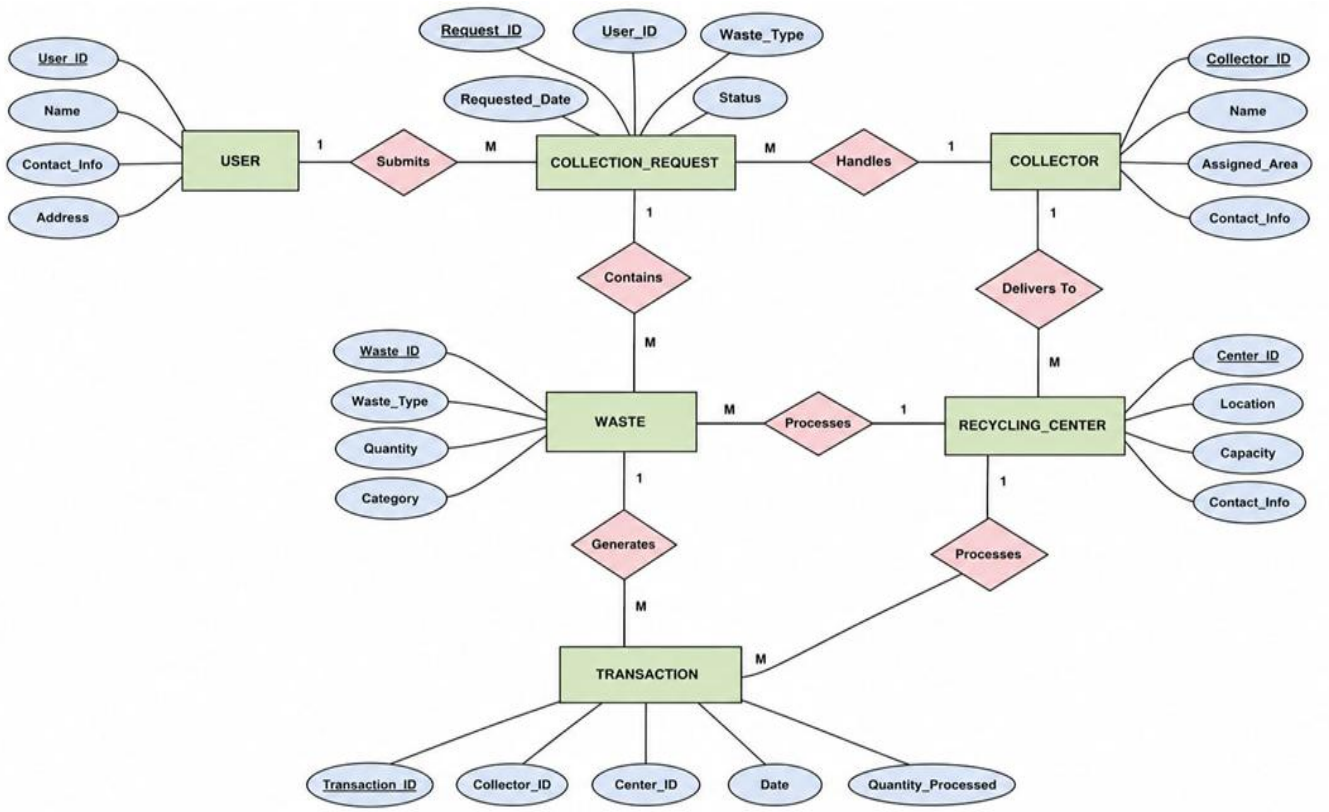


Figure 1: ER Diagram of Recycle Management System

5. RELATIONAL SCHEMA

The relational schema for the database is as follows:

Relation	Relational Schema
USER	User_ID (PK), Name, Contact_Info, Address
COLLECTOR	Collector_ID (PK), Name, Assigned_Area, Contact_Info
COLLECTION_REQUEST	Request_ID (PK), User_ID (FK), Collector_ID (FK), Waste_Type, Requested_Date, Status
WASTE	Waste_ID (PK), Waste_Type, Quantity, Category, Request_ID (FK), Center_ID (FK)
RECYCLING_CENTER	Center_ID (PK), Location, Capacity, Contact_Info
TRANSACTION	Transaction_ID (PK), Collector_ID (FK), Center_ID (FK), Date, Quantity_Processed, Waste_ID (FK)

6. TABLE DESIGN

6.1 User Table

Attribute	Data Type	Constraint	Description
User_ID	INT	PK, NOT NULL	Unique user identifier
Name	VARCHAR(100)	NOT NULL	User name
Contact_Info	VARCHAR(100)	NOT NULL	Contact information
Address	VARCHAR(255)	—	User address

6.2 Collector Table

Attribute	Data Type	Constraint	Description
Collector_ID	INT	PK, NOT NULL	Unique collector identifier
Name	VARCHAR(100)	NOT NULL	Collector name
Assigned_Area	VARCHAR(100)	—	Assigned collection area
Contact_Info	VARCHAR(100)	NOT NULL	Collector contact information

6.3 Collection Request Table

Attribute	Data Type	Constraint	Description
Request_ID	INT	PK, NOT NULL	Unique request identifier
User_ID	INT	FK	References USER
Waste_Type	VARCHAR(50)	—	Type of waste requested
Requested_Date	DATE	—	Date of request

6.4 Waste Table

Attribute	Data Type	Constraint	Description
Waste_ID	INT	PK, NOT NULL	Unique waste identifier
Waste_Type	VARCHAR(50)	—	Type of waste
Quantity	DECIMAL(10,2)	—	Quantity of waste
Category	VARCHAR(50)	—	Waste category
Request_ID	INT	FK	References COLLECTION_REQUEST

6.5 Recycling Center Table

Attribute	Data Type	Constraint	Description
Center_ID	INT	PK, NOT NULL	Unique center identifier
Location	VARCHAR(150)	—	Center location
Capacity	DECIMAL(10,2)	CHECK (Capacity >= 0)	Center capacity
Contact_Info	VARCHAR(100)	—	Center contact information
Status	VARCHAR	CHECK	Available or Assigned

6.6 Transaction Table

Attribute	Data Type	Constraint	Description
Transaction_ID	INT	PK, NOT NULL	Unique transaction identifier
Collector_ID	INT	FK	References Collector
Center_ID	INT	FK	References Collection_Center
Date	DATE	—	Transaction date

6.7 Keys and Constraints

Constraint Type	Applied To / Purpose
Primary Key	User_ID, Collector_ID, Request_ID, Waste_ID, Center_ID, Transaction_ID
Foreign Key	Collection_Request.User_ID, Waste.Request_ID, Transaction.Collector_ID, Transaction.Center_ID
NOT NULL	Required fields such as IDs and names.
CHECK	Capacity >= 0 and Quantity_Processed >= 0, where applicable.

7. NORMALIZATION

Normalization organizes data into related relations to reduce redundancy and update anomalies. The database design follows normalization up to Third Normal Form (3NF).

7.1 Unnormalized Form (UNF)

A single record may contain user details, waste information, collection request details, collector details, recycling center details, and transaction information. Repeated information can create data redundancy.

7.2 First Normal Form (1NF)

In 1NF, repeating groups are removed, and each attribute contains a single value. Each record represents one entry in the respective table.

7.3 Second Normal Form (2NF)

In 2NF, partial dependencies on a composite primary key are removed. Each non-key attribute depends on the entire primary key.

7.4 Third Normal Form (3NF)

In 3NF, transitive dependencies are removed. User, collector, collection request, waste, recycling center, and transaction details are stored in separate tables and connected through foreign keys.

Relation	Reason for Separation
USER	Stores user details.
COLLECTOR	Stores collector details.
COLLECTION_REQUEST	Stores collection request information.
WASTE	Stores waste type, quantity, and category.
RECYCLING_CENTER	Stores recycling center details.

8. SQL IMPLEMENTATION

8.1 DDL – Data Definition Language

The following DDL is a documentation-ready implementation aligned with the relational schema. Data types may be adjusted to the specific DBMS used by the student team.

```
-- Create Database
CREATE DATABASE recycling_management;
USE recycling_management;

-- Create User  Table
CREATE TABLE Users (
   user_id INT PRIMARY KEY AUTO_INCREMENT,
   name VARCHAR(100) NOT NULL,
   contact_info VARCHAR(100) NOT NULL UNIQUE,
   address VARCHAR(255) NOT NULL
);

-- Create Collector Table
CREATE TABLE Collector (
   collector_id INT PRIMARY KEY AUTO_INCREMENT,
   name VARCHAR(100) NOT NULL,
   assigned_area VARCHAR(100) NOT NULL,
   contact_info VARCHAR(100) NOT NULL UNIQUE
);

-- Create Recycling_Center Table
CREATE TABLE Recycling_Center (
   center_id INT PRIMARY KEY AUTO_INCREMENT,
   location VARCHAR(150) NOT NULL UNIQUE,
   capacity DECIMAL(10,2) NOT NULL,
   contact_info VARCHAR(100) NOT NULL
);
```

```
-- Create Collection_Request Table
CREATE TABLE Collection_Request (
    request_id INT PRIMARY KEY AUTO_INCREMENT,
    user_id INT NOT NULL,
    collector_id INT,
    waste_type VARCHAR(50) NOT NULL,
    requested_date DATE NOT NULL,
    status ENUM('Pending','Assigned','Collected','Completed','Cancelled')
        DEFAULT 'Pending',
    FOREIGN KEY (user_id)
        REFERENCES Users(user_id)
        ON DELETE CASCADE,
    FOREIGN KEY (collector_id)
        REFERENCES Collector(collector_id)
        ON DELETE SET NULL
);

-- Create Waste Table
CREATE TABLE Waste (
    waste_id INT PRIMARY KEY AUTO_INCREMENT,
    request_id INT NOT NULL,
    center_id INT,
    waste_type VARCHAR(50) NOT NULL,
    quantity DECIMAL(10,2) NOT NULL,
    category ENUM('Recyclable','Reusable','Hazardous','Organic') NOT NULL,
    FOREIGN KEY (request_id)
        REFERENCES Collection_Request(request_id)
        ON DELETE CASCADE,
    FOREIGN KEY (center_id)
        REFERENCES Recycling_Center(center_id)
        ON DELETE SET NULL
);
```

```
-- Create Transactions Table
CREATE TABLE Transactions (
   transaction_id INT PRIMARY KEY AUTO_INCREMENT,
   waste_id INT NOT NULL,
   collector_id INT NOT NULL,
   center_id INT NOT NULL,
   transaction_date DATE NOT NULL,
   quantity_processed DECIMAL(10,2) NOT NULL,

   FOREIGN KEY (waste_id)
      REFERENCES Waste(waste_id),
   FOREIGN KEY (collector_id)
      REFERENCES Collector(collector_id),
   FOREIGN KEY (center_id)
      REFERENCES Recycling_Center(center_id)
);
```

8.2 DML – Data Manipulation Language

```
--Insert into Users
INSERT INTO Users (name, contact_info, address) VALUES
('Ravi Kumar', '9876543210', 'Vijayawada'),
('Priya Sharma', '9876543211', 'Guntur'),
('Arjun Reddy', '9876543212', 'Hyderabad'),
('Sneha Rao', '9876543213', 'Rajahmundry'),
('Kiran Das', '9876543214', 'Visakhapatnam'),
('Manoj', '9876543215', 'Eluru'),
('Divya', '9876543216', 'Kakinada'),
('Pavan', '9876543217', 'Ongole');

--Insert into Collector
INSERT INTO Collector (name, assigned_area, contact_info) VALUES
('Ramesh', 'Vijayawada', '9000000001'),
('Suresh', 'Guntur', '9000000002'),
('Mahesh', 'Hyderabad', '9000000003'),
```

('Karthik', 'Eluru', '9000000004'),
('Naveen', 'Kakinada', '9000000005');

--Insert into Recycling_Center
INSERT INTO Recycling_Center
(location, capacity, contact_info) VALUES
('Vijayawada Center', 1000.00, '8000000001'),
('Guntur Center', 800.00, '8000000002'),
('Hyderabad Center', 1500.00, '8000000003'),
('Eluru Center', 900.00, '8000000004'),
('Kakinada Center', 1100.00, '8000000005');

Insert into Collection_Request

INSERT INTO Collection_Request
(user_id, collector_id, waste_type, requested_date, status) VALUES
(1, 1, 'Plastic', '2026-09-01', 'Completed'),
(2, 2, 'Paper', '2026-09-02', 'Collected'),
(3, 3, 'Metal', '2026-09-03', 'Assigned'),
(4, NULL, 'Glass', '2026-09-04', 'Pending'),
(5, 1, 'Plastic', '2026-09-05', 'Completed'),
(6, 4, 'Paper', '2026-09-10', 'Completed'),
(7, 5, 'Glass', '2026-09-11', 'Pending'),
(8, 4, 'Metal', '2026-09-12', 'Assigned');

--Insert into Waste
INSERT INTO Waste
(request_id, center_id, waste_type, quantity, category) VALUES
(1, 1, 'Plastic Bottles', 50.00, 'Recyclable'),
(2, 2, 'Paper Waste', 30.00, 'Reusable'),
(3, 3, 'Metal Cans', 20.00, 'Recyclable'),
(4, NULL, 'Glass Bottles', 15.00, 'Recyclable'),
(5, 1, 'Plastic Containers', 40.00, 'Recyclable'),
(6, 4, 'Cardboard', 25.00, 'Reusable'),
(7, NULL, 'Glass Bottles', 18.00, 'Recyclable'),
(8, 5, 'Aluminum Cans', 35.00, 'Recyclable');

```
--Insert into Transactions
INSERT INTO Transactions
(waste_id, collector_id, center_id,
 transaction_date, quantity_processed) VALUES
(1, 1, 1, '2026-09-06', 40.00),
(2, 2, 2, '2026-09-07', 25.00),
(3, 3, 3, '2026-09-08', 15.00),
(5, 1, 1, '2026-09-09', 30.00),
(6, 4, 4, '2026-09-13', 20.00),
(8, 4, 5, '2026-09-14', 30.00);
```

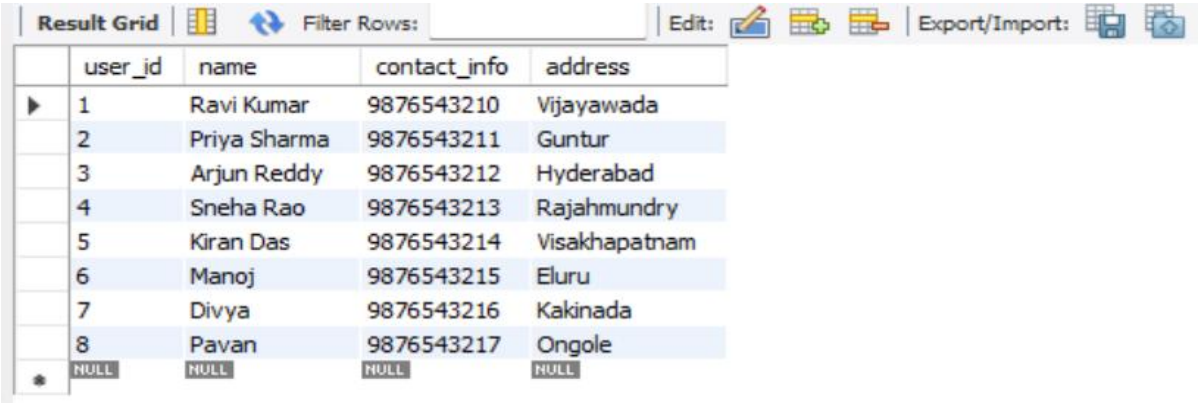
9. SQL QUERIES & SCREENSHOTS

The following SQL queries demonstrate data retrieval, filtering, joins, aggregation, and reporting operations in the Recycling Management System.

Q1. Display all users

SELECT * FROM Users;

Output:

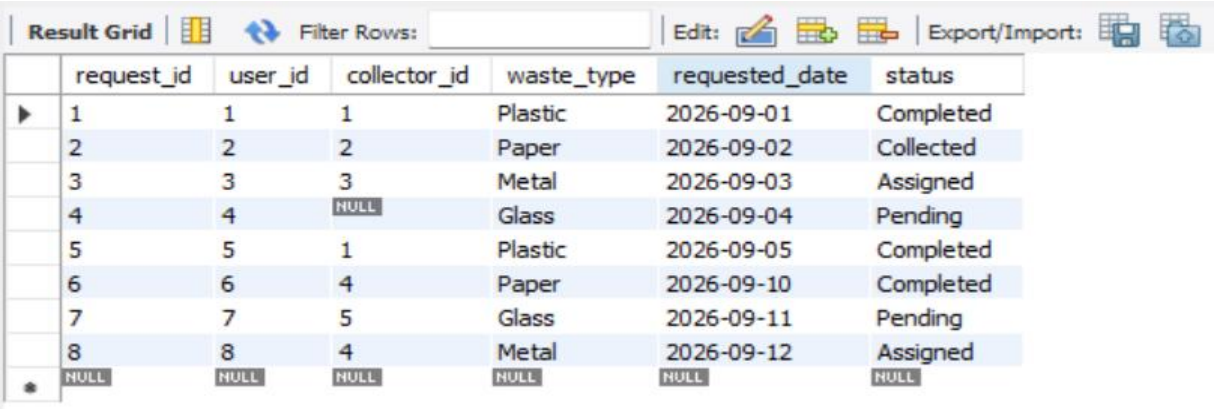


user_id	name	contact_info	address
1	Ravi Kumar	9876543210	Vijayawada
2	Priya Sharma	9876543211	Guntur
3	Arjun Reddy	9876543212	Hyderabad
4	Sneha Rao	9876543213	Rajahmundry
5	Kiran Das	9876543214	Visakhapatnam
6	Manoj	9876543215	Eluru
7	Divya	9876543216	Kakinada
8	Pavan	9876543217	Ongole
NULL	NULL	NULL	NULL

Q2. Display all collection requests

SELECT * FROM Collection_Request;

Output:

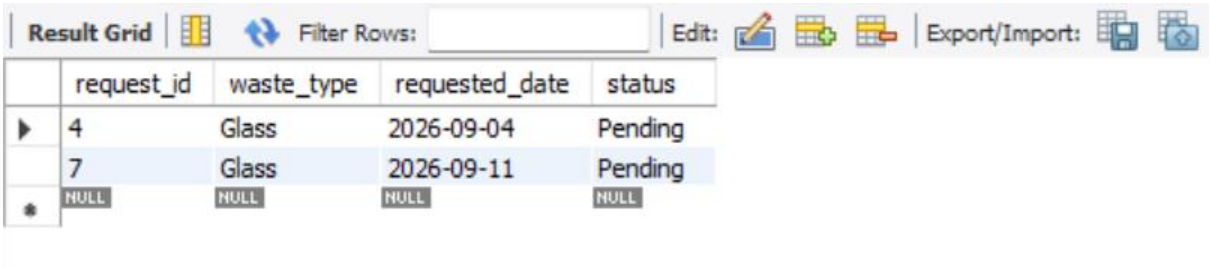


request_id	user_id	collector_id	waste_type	requested_date	status
1	1	1	Plastic	2026-09-01	Completed
2	2	2	Paper	2026-09-02	Collected
3	3	3	Metal	2026-09-03	Assigned
4	4	NULL	Glass	2026-09-04	Pending
5	5	1	Plastic	2026-09-05	Completed
6	6	4	Paper	2026-09-10	Completed
7	7	5	Glass	2026-09-11	Pending
8	8	4	Metal	2026-09-12	Assigned
NULL	NULL	NULL	NULL	NULL	NULL

Q3. Display pending collection requests

```
SELECT request_id, waste_type, requested_date, status
FROM Collection_Request
WHERE status = 'Pending';
```

Output:

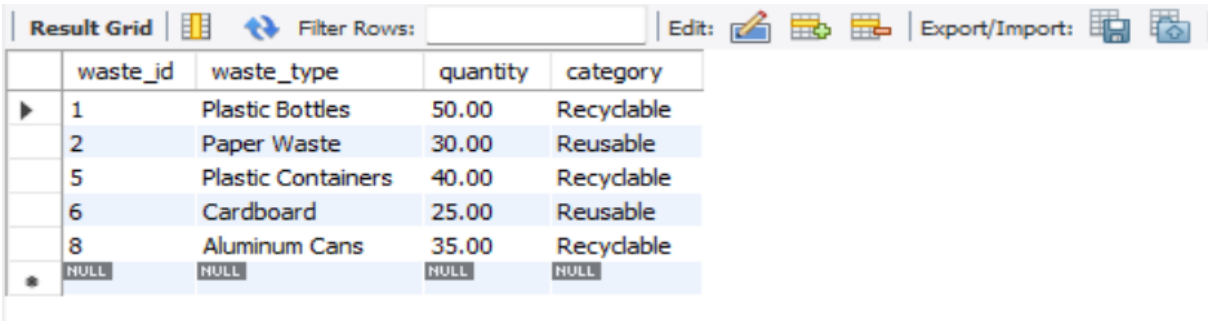


request_id	waste_type	requested_date	status
4	Glass	2026-09-04	Pending
7	Glass	2026-09-11	Pending
NULL	NULL	NULL	NULL

Q4. Display waste with quantity greater than 20

```
SELECT waste_id, waste_type, quantity, category
FROM Waste
WHERE quantity > 20;
```

Output:

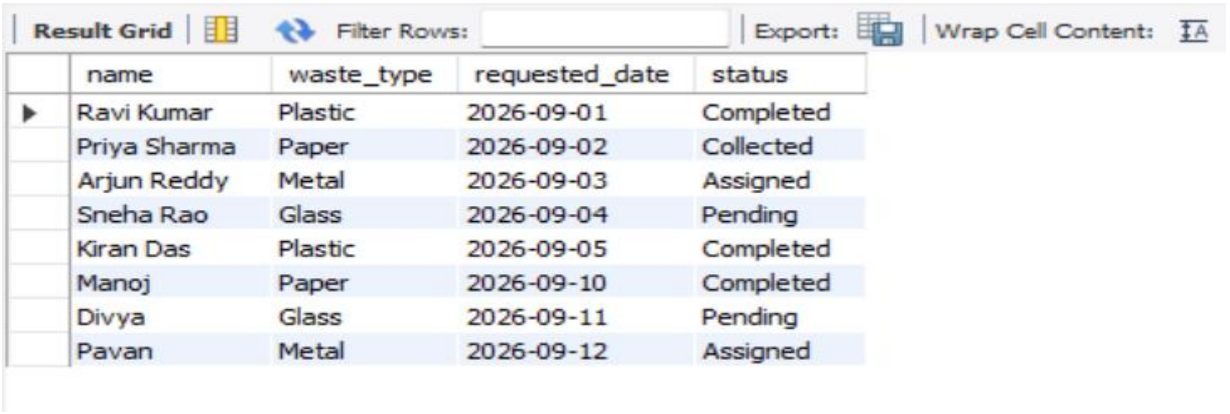


waste_id	waste_type	quantity	category
1	Plastic Bottles	50.00	Recyclable
2	Paper Waste	30.00	Reusable
5	Plastic Containers	40.00	Recyclable
6	Cardboard	25.00	Reusable
8	Aluminum Cans	35.00	Recyclable
NULL	NULL	NULL	NULL

Q5. Display collection requests with user names

```
SELECT u.name, r.waste_type,
       r.requested_date, r.status
FROM Users u
JOIN Collection_Request r
ON u.user_id = r.user_id;
```

Output:

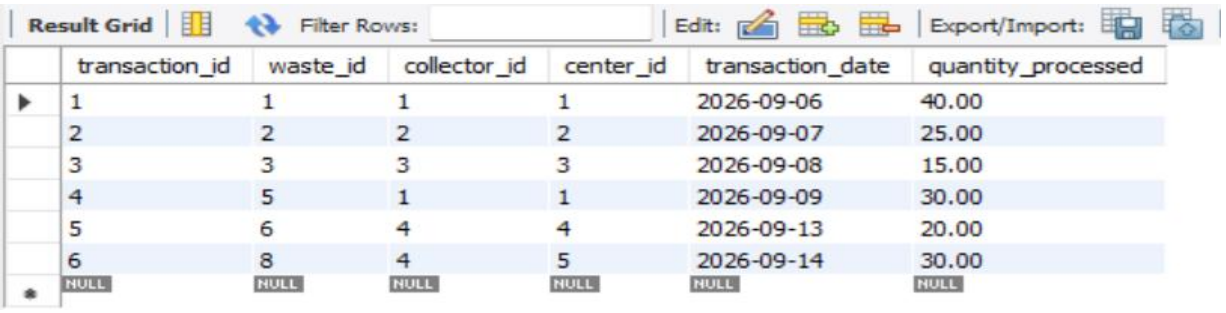


name	waste_type	requested_date	status
Ravi Kumar	Plastic	2026-09-01	Completed
Priya Sharma	Paper	2026-09-02	Collected
Arjun Reddy	Metal	2026-09-03	Assigned
Sneha Rao	Glass	2026-09-04	Pending
Kiran Das	Plastic	2026-09-05	Completed
Manoj	Paper	2026-09-10	Completed
Divya	Glass	2026-09-11	Pending
Pavan	Metal	2026-09-12	Assigned

Q6. Display all recycling transactions

```
SELECT * FROM Transactions;
```

Output:

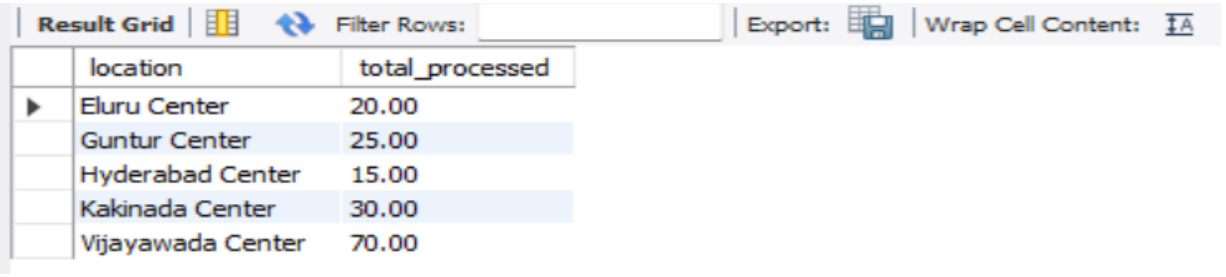


transaction_id	waste_id	collector_id	center_id	transaction_date	quantity_processed
1	1	1	1	2026-09-06	40.00
2	2	2	2	2026-09-07	25.00
3	3	3	3	2026-09-08	15.00
4	5	1	1	2026-09-09	30.00
5	6	4	4	2026-09-13	20.00
6	8	4	5	2026-09-14	30.00
NULL	NULL	NULL	NULL	NULL	NULL

Q7. Calculate processed quantity for each recycling center

```
SELECT rc.location,
    SUM(t.quantity_processed) AS total_processed
FROM Recycling_Center rc
JOIN Transactions t
ON rc.center_id = t.center_id
GROUP BY rc.center_id, rc.location;
```

Output:

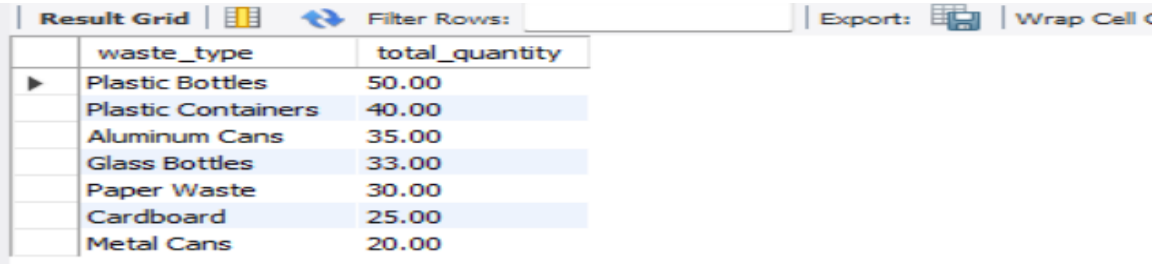


location	total_processed
Eluru Center	20.00
Guntur Center	25.00
Hyderabad Center	15.00
Kakinada Center	30.00
Vijayawada Center	70.00

Q8. Display total waste quantity by type

```
SELECT waste_type,
    SUM(quantity) AS total_quantity
FROM Waste
GROUP BY waste_type
ORDER BY total_quantity DESC;
```

Output:



waste_type	total_quantity
Plastic Bottles	50.00
Plastic Containers	40.00
Aluminum Cans	35.00
Glass Bottles	33.00
Paper Waste	30.00
Cardboard	25.00
Metal Cans	20.00

16. Results & Analysis

16.1 Key Insights

The database implementation successfully centralizes recyclable waste information and provides the following insights:

- **Collection Statistics:** Track the number and status of collection requests.
- **Waste Analysis:** Identify waste types and calculate total waste quantities.
- **Collector Performance:** Analyze the number of collection requests handled by collectors.
- **Recycling Center Usage:** Summarize processed waste quantities at each recycling center.

16.2 Performance Observations

- **Fast Retrieval:** Indexes help improve the efficiency of frequently used queries.
- **Data Integrity:** Primary keys and foreign keys maintain valid relationships between tables.
- **Quantity Validation:** A trigger prevents the total processed quantity from exceeding the recorded waste quantity.
- **Simplified Reporting:** Views provide organized information about collection requests, recycling center usage, and collector activity.
- **Scalability:** The database design supports adding more users, collectors, collection requests, waste records, and recycling transactions.

16.3 Recycling Transaction Summary

Transaction_ID	Waste_ID	**Collector_ID**	**Center_ID**	**Quantity_Processed**	**Date**
1	1	1	1	40.00	2026-09-06
2	2	2	2	25.00	2026-09-07
3	3	3	3	15.00	2026-09-08
4	5	1	1	30.00	2026-09-09
5	6	4	4	20.00	2026-09-13
6	8	4	5	30.00	2026-09-14

Summary:

- **Collection Status:** Completed, Collected, Assigned, Pending, and Cancelled.
- **Waste Processing:** The system tracks waste quantities and processed quantities across recycling centers.
- **Foreign Key Constraints:** Maintain valid relationships between users, collectors, collection requests, waste records, and transactions.
- **Trigger Validation:** Prevents the total processed quantity from exceeding the recorded waste quantity.

17. Conclusion & Future Enhancements

17.1 Conclusion

The Recycling Management System provides a structured relational database solution for managing users, collectors, collection requests, waste materials, recycling centers, and transactions. The database uses primary keys, foreign keys, and constraints to maintain data integrity and consistency. Normalization helps reduce data redundancy and update anomalies. The MySQL implementation demonstrates data insertion, retrieval, updating, deletion, and reporting through SQL queries. The project provides a foundation for efficient waste collection tracking and recycling management.

17.2 Future Enhancements

- Real-time tracking of waste collection vehicles.
- Mobile application for users and collectors.
- Automated notifications for collection requests.
- Dashboards for recycling statistics and reports.
- QR codes for waste tracking and identification.
- Integration with IoT-enabled smart waste bins.
- Cloud database integration for improved accessibility.
- Advanced analytics for monitoring recycling performance.

18. References

- Korth, H. F., Silberschatz, A., and Sudarshan, S., Database System Concepts, McGraw Hill.
- Elmasri, R. and Navathe, S. B., Fundamentals of Database Systems, Pearson.
- MySQL Documentation, Oracle Corporation.
- Aditya University Project Guidelines and Templates.